Economics Letters 258 (2026) 112748

Contents lists available at ScienceDirect

Economics Letters

journal homepage: www.elsevier.com/locate/ecolet

Climate physical risks and the vulnerability of global agricultural commodities

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ARTICLE INFO

JEL classification:
C32
Q13
Q54

Keywords:
Climate physical risk
Agricultural commodity
Vulnerability
Food security

ABSTRACT

This paper focuses on the impacts of climate physical risks on crop prices and proposes a novel method to evaluate the vulnerability of five agricultural commodities, i.e., wheat, maize, soybean, rice, and barley. We identified the high vulnerability of maize and high resistance of rice to shocks of physical risks. During the period from 2020 to 2022, the vulnerability of global agricultural commodity markets sharply rose. The findings provide novel insight into the influence of climate change on agricultural sectors.

1. Introduction

Climate risks pose a great challenge to the world (Guo et al., 2023). Agriculture, a highly sensitive sector to climate change, is suffering from these threats. Extreme weather and various natural disasters become more often and severe worldwide, like severe droughts caused by El Niño in Southeast Asia in February 2020, the impressed locust plague spanning East Africa, West Asia, and South Asia from January 2020 to June 2022, etc., undoubtedly exerting huge impacts on the global food security (Cui and Zhong, 2024; Wang et al., 2024). The State of Food Security and Nutrition in the World (2024) claims a sharp rise in global hunger from 2019, which has persisted at nearly the same high level in recent years.

From the perspective of economics and finance, climate risks can be categorized into two types, i.e., transition and physical risks (Arteaga-Garavito et al., 2025; Battiston et al., 2021). Climate physical risks (CPR) exacerbate global food insecurity primarily through the following four channels (Hasegawa et al., 2018; Jägermeyr et al., 2021): Firstly, *supply reduction*: Extreme weather events (e.g., droughts, floods) directly damage yields in major producing regions, causing global shortages. Secondly, *inventory depletion*: Frequent disasters compel nations to draw down strategic reserves, diminishing their price stabilization function. Thirdly, *trade distortions*: Export restrictions (e.g., bans) implemented by key suppliers distort trade flows, triggering supply chain disruptions and amplifying regional supply-demand imbalances. Finally, *demand inelasticity*: The inherent inelasticity of food demand means even minor supply shocks can induce significant price volatility. Collectively, these mechanisms transmit climate-induced supply chain disruptions and trade interventions to global markets, driving up prices and magnifying price volatility. Hence, differing from the “climate resilience”, “production vulnerability”, and “system vulnerability” in existing research (Cui and Zhong, 2024; Fang et al., 2024; Li et al., 2025; Wang et al., 2024), the impact of CPR on the price dynamics of agricultural products is the ultimate manifestation provoked by climate change, which naturally brings about the key issue about “price vulnerability”, an important yet not very well-explored question.

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https://doi.org/10.1016/j.econlet.2025.112748
Received 3 June 2025; Received in revised form 16 November 2025; Accepted 23 November 2025
Available online 24 November 2025
0165-1765/

This paper aims to propose a quantitative framework to capture the price vulnerability of global agricultural commodities geared at CPR. We consider five agricultural commodities, i.e., wheat, maize, soybean, rice, and barley, which are constituents of the IGC[4] Grains and Oilseeds Index and are also vital crops for food sources. Drawing on the portfolio theory approach, we further aggregate all vulnerability sub-indices into the composite indicator to reflect the whole price susceptibility, which delineates an overview for policy-makers to know the exact level of food crisis.

The key contributions are mainly twofold: Academically, this study pioneers a quantitative impulse response-based framework to assess the price vulnerability of global agricultural commodities to CPR, revealing crop-specific heterogeneity and temporal crisis patterns overlooked in prior aggregated analyses. Practically, it delivers actionable policy pathways—including targeted crop resilience investments, strategic reserve management, and trade coordination—enabling evidence-based interventions during vulnerability peaks. The framework empowers policymakers to preemptively mitigate food security threats under evolving climate risks.

2. Data and methodology

2.1. Data

The daily indices of wheat, maize, soybean, rice, and barley are obtained from IGC. These price indices, derived from export quotations of multiple varieties from the major producers (countries or regions), can reflect the historical trends in international agricultural spot prices well. As regards the measures of climate risks, we collect the news-based indices constructed by Faccini et al. (2023), wherein they utilized textual and narrative analysis to get four types of climate-related risks, namely global warming (GW), natural disasters (ND), international summits, and U.S. climate policy. They used the Latent Dirichlet Allocation (LDA) to group the heterogeneous news (published in Reuters) into the above-mentioned four climate-subcategories, and the estimated topic share for any given topic is the proportion of words contained in the article. The first two indices (viz. GW and ND) are associated with CPR, and the other two indices measure transition risks. We transform all daily data into monthly data via averaging, and the data sample spans from January 2000 to January 2025. For stationarity, all series undergo log differencing as follows:

$$r_t = 100 \times \ln(p_t / p_{t-1}) \tag{1}$$

Table 1 reports descriptive statistics. All indicators show non-normality according to the JB test statistics and are stationary based on the ADF and ERS statistics. In terms of kurtosis, rice and barley showcase prominently more leptokurtic and fat-tail features than other crops. Furthermore, among the five agricultural commodities, rice is the least volatile.

2.2. Theoretical framework of vulnerability

We simulate the response pathway of system disturbance after suffering the shock in Fig. 1, which can help us elaborate upon the vulnerability characteristics of the system. The system undergoes a shock at t_0, generates disturbance that causes a deviation from the normal state, and recovers to the original state at t_N. Considering two fictitious scenes shown in Fig. 1, namely Scene 1 and Scene 2, they show pronouncedly tangible differences in terms of response pathways. Intuitively speaking, the stability of Scene 1 collapses faster after being impacted, and the system disturbance reaches its peak faster. Hence, we think Scene 1 is more vulnerable than Scene 2. To differentiate and reflect this feature, we define the vulnerability metric as:

$$vul = \sum_{n=1}^{N} \phi_n \frac{N-n+1}{\sum_{n=1}^{N} n} \tag{2}$$

where ϕ_n is the response value at t_n, $n = 1, 2, \cdots, N$, and N is the response horizon. This weighted average method not only considers the disturbance magnitude but also takes into account the deterioration speed of the system stability. As a result, even if the system disturbances have the same duration $t_N - t_0$ and $Peak_1 = Peak_2$, the vulnerability of Scene 1 is still greater than that of Scene 2, i.e., $vul_1 > vul_2$.

2.3. Measurement methods of vulnerability

To calculate the vulnerability of agricultural commodities to CPR, we first use the dynamic factor model to extract the latent common factor from two physical-risk-related indices (i.e., GW and ND) as the CPR. Then, inspired by the studies of Chen and Sun (2024) and Yang et al. (2025), who used the time-varying impulse response function derived from the time-varying parameter vector auto-regression (TVP-VAR) to capture response dynamics, we apply the absolute value of impulse response function (denoted as $|IRF|$) to replace ϕ_n in Eq. (2):

$$vul_t = \sum_{n=1}^{N} \left| IRF_{t,n}^{crop \leftarrow CPR} \right| \frac{N-n+1}{\sum_{n=1}^{N} n} \tag{3}$$

where $IRF_{t,n}^{crop \leftarrow CPR}$ represents the response of agricultural commodities to the shock of CPR at time t. N is the response horizon when $|IRF|$ converges to zero. For more technical details about TVP-VAR, please refer to Primiceri (2005) and Nakajima et al. (2011).

2.4. Aggregation of composite vulnerability index

We utilize the portfolio theory to aggregate the vulnerability measure of every agricultural commodity (Duprey et al., 2017; Louzis and Vouldis, 2012). This method can reflect the relative importance of each sub-index via incorporating the weight factor and can also capture the co-movement interrelationship among all sub-indices. The composite vulnerability index is calculated as follows:

$$Vul_t = \sqrt{(W \odot \widetilde{Vul}_t)' \Omega_t (W \odot \widetilde{Vul}_t)} \tag{4}$$

where $\odot$ is the Hadamard product, W is the weight vector, and $\widetilde{Vul}_t = \left(Vul_t^{wheat}, Vul_t^{maize}, Vul_t^{soybean}, Vul_t^{rice}, Vul_t^{barley} \right)$. Ω_t is the 5×5 time-varying cross-correlation matrix of returns of five agricultural commodities:

$$\Omega_t = \begin{bmatrix} 1 & \rho_{12,t} & \rho_{13,t} & \rho_{14,t} & \rho_{15,t} \\ \rho_{21,t} & 1 & \rho_{23,t} & \rho_{24,t} & \rho_{25,t} \\ \rho_{31,t} & \rho_{32,t} & 1 & \rho_{34,t} & \rho_{35,t} \\ \rho_{41,t} & \rho_{42,t} & \rho_{43,t} & 1 & \rho_{45,t} \\ \rho_{51,t} & \rho_{52,t} & \rho_{53,t} & \rho_{54,t} & 1 \end{bmatrix} \tag{5}$$

The time-varying cross-correlation coefficients $\rho_{ij,t}$ are estimated by an exponentially weighted moving average (EWMA) specification[5] with smoothing parameter $\lambda = 0.93$.[6]

[4] IGC: International Grains Council. https://www.igc.int

[5] The conditional covariance matrix derived from multivariable GARCH models (like the DCC or BEKK specification) can be also chosen. We showcase the composite indicator of vulnerability with Ω_t estimated by DCC-GARCH and make comparisons in Fig. A2 in the Appendix.

[6] In fact, calculation results are insensitive to the choice of smoothing parameter. For robustness, we try $\lambda \in \{0.85, 0.87, 0.89, 0.91, 0.93, 0.95\}$ and obtain very similar results, which are not displayed due to space limitations.

Table 1
Descriptive statistics of log differences of all series.

	Wheat	Maize	Soybean	Rice	Barley	GW	ND
Mean	0.231	0.284	0.238	0.233	0.269	0.509	1.014
Max	23.263	18.848	16.193	31.406	41.518	191.171	266.128
Min	−18.590	−21.626	−21.302	−14.052	−22.708	−195.368	−197.005
Var	25.776	31.503	25.719	16.606	32.083	2574.949	3411.457
Skew	0.736	0.070	0.074	2.448	1.492	0.074	0.286
Kurt	6.558	4.586	4.222	20.224	14.348	3.944	4.602
JB	185.382***	31.682***	18.940***	4007.883***	1721.047***	11.405***	36.180***
ADF	−5.957***	−6.473***	−6.513***	−7.138***	−6.410***	−9.666***	−9.887***
ERS	−5.768***	−6.997***	−6.036***	−7.234***	−5.217***	−3.702***	−1.904*

Notes: The table reports the descriptive statistics including mean (Mean), maximum (Max), minimum (Min), variance (Var), skewness (Skew), kurtosis (Kurt), the Jarque-Bera statistics of the normality test (JB), the ADF unit root test (ADF), and the Elliott–Rothenberg–Stock statistics of the stationarity test (ERS). ***, **, and * denote the rejections of the null hypothesis at the significance levels of 1 %, 5 %, and 10 %, respectively.

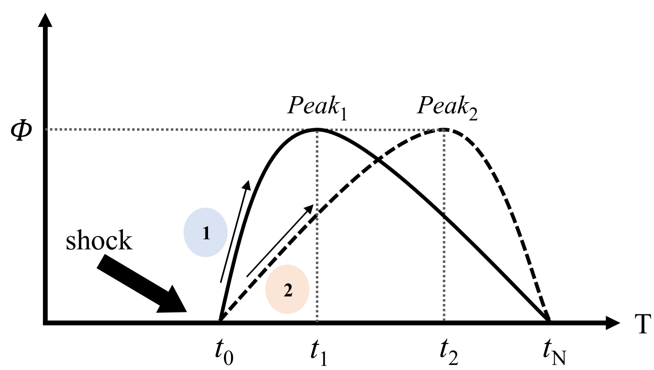


Fig. 1. Two forms of system disturbances when confronted with shocks.

3. Empirical results

3.1. Time-varying impulse response functions to CPR

Fig. 2 showcases all trends of agricultural commodity prices and two types of climate physical risks. Two measures of physical risks share some similar historical trajectories. This is easy to understand, as with the worsening of global warming, many natural disasters also become more frequent. Prices of agricultural commodities soared mainly during three periods, namely the global food crisis mainly during 2005–2008, a new round of prominent uptrend starting from 2010, and the recent sharp rise during 2020–2022. Of special note is that physical risks remained at a high level only during the two aforementioned periods, not including the period of 2010–2013. This offers a good reference for us to verify the rationality of future empirical results.

After using the dynamic factor model to extract the latent common factor of two types of physical risks, we compute all time-varying impulse response functions of agricultural commodities to CPR, as shown in Fig. 3. CPR mainly has a markedly positive impact on crop prices, and the strength of *IRF* generally converges to zero at approximately 5 months in terms of the period dimension. Therefore, we set the response horizon $N = 5$ in Eq. (3) and also test $N = 4,\ 6$ for robustness, the results of which are reported in Fig. A1 in the Appendix. The results of $N = 4,\ 5,\ 6$ are almost identical in terms of evolutionary trends.

3.2. Vulnerability characteristics of agricultural commodities

Fig. 4 illustrates the time-varying dynamics and statistical characteristics of the vulnerability of five crops. From the temporal lens, wheat, maize, soybean, and barley showed sharply rising vulnerability to the impacts of CPR during 2005–2008 and 2020–2022. However, during the second food crisis (2010–2013), their vulnerability was kept at a low level. Recalling low physical risks during this period in Fig. 2, it turns out that our vulnerability measure is effective and sensitive. The vulnerability trend of rice was a little different and has continuously deteriorated recently.

Compared with each other, Fig. 4(b) indicates that maize is the most susceptible to the shock of CPR, and soybean also shows great vulnerability. Next are barley and wheat, whose vulnerability is moderate. Rice has the strongest resistance and is the least vulnerable. The resilience of global rice price stems from a "systematic buffer" formed by its unique geographical distribution (dominated by Asia and relatively decentralized), robust infrastructure (mainly well-developed irrigation system), a vast national reserve system spurred by its critical strategic importance (particularly in China and India—the two largest producers and consumers—which maintain exceptionally high levels of strategic rice re-

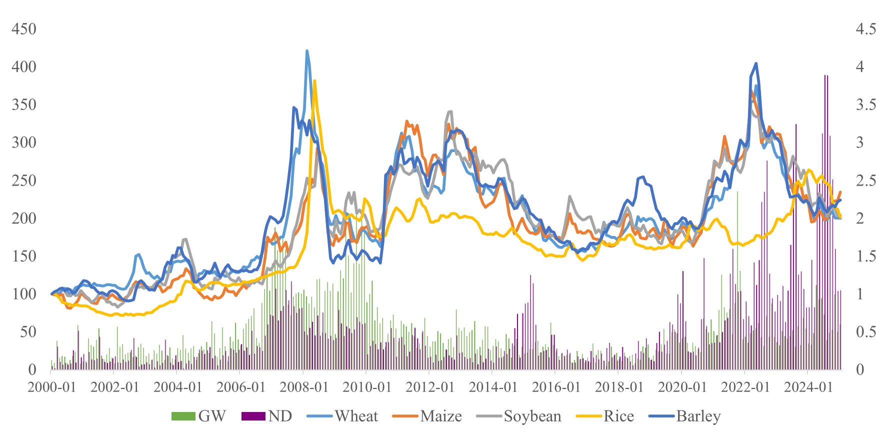


Fig. 2. Trends of climate physical risks and agricultural commodity indices.

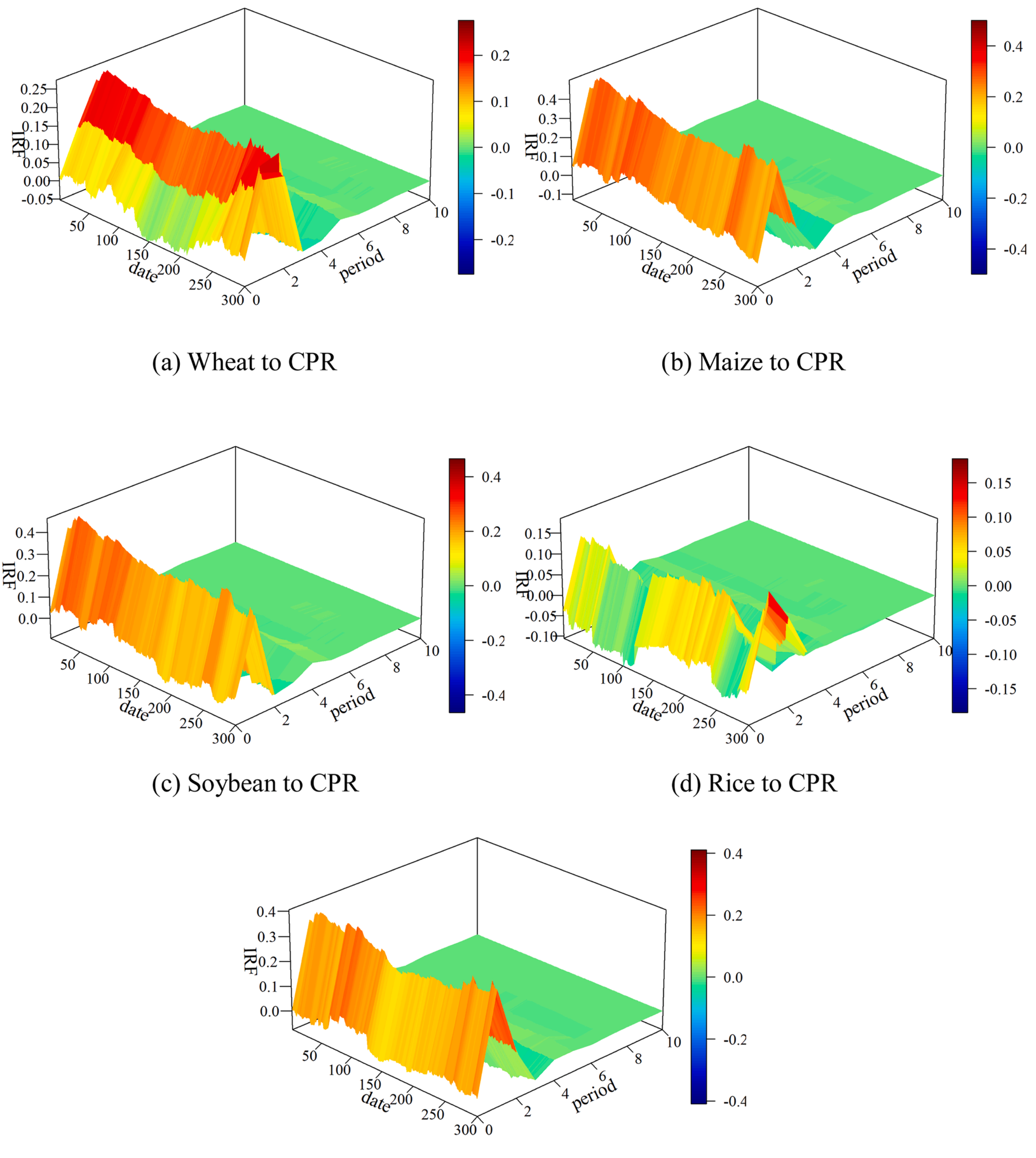
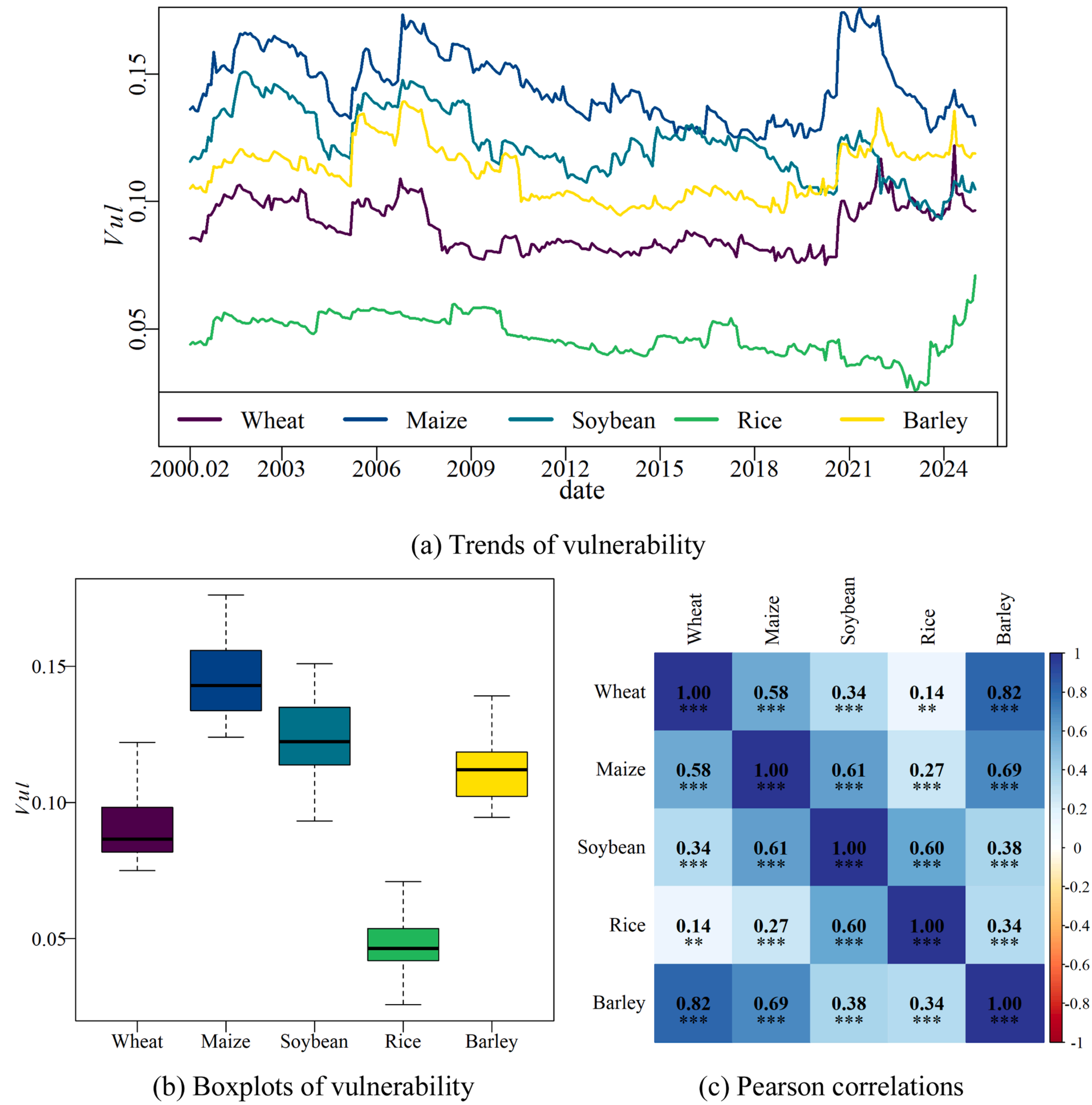


Fig. 3. All time-varying impulse response functions to CPR.



(a) Trends of vulnerability

(b) Boxplots of vulnerability

(c) Pearson correlations

Fig. 4. Vulnerability metrics of five agricultural commodities.

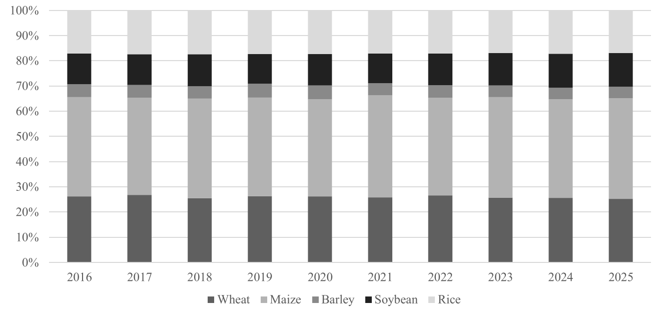


Fig. 5. Production proportions. The results are calculated based on the total production of these five grains and oilseeds. Data source: IGC.

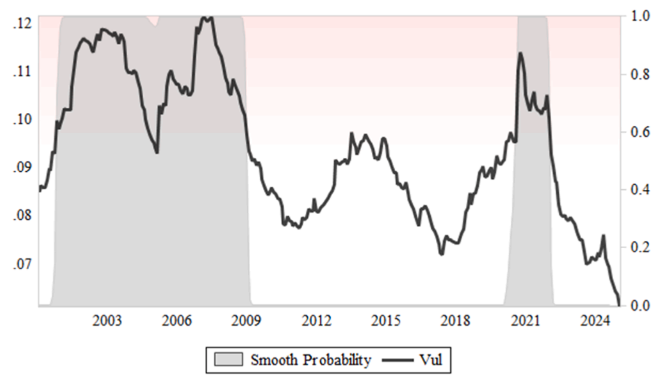


Fig. 6. Composite indicator of vulnerability and smooth probabilities of high-vulnerability states.

serves), and a highly policy-intervened international trade structure. In contrast, maize and soybean exhibit extreme sensitivity and vulnerability to climate-related physical risks in core production regions due to their highly concentrated production (with exports nearly monopolized by the U.S., Brazil, and Argentina), reliance on rain-fed agriculture (heavily dependent on natural precipitation), and a trade system that is deeply market-driven and globalized.

From the perspective of correlations (in Fig. 4(c)), vulnerability changes of wheat and barley are highly positively correlated (0.82), whilst those of wheat and rice are weakly positively correlated (0.14).

3.3. Composite index of vulnerability

We use production proportions as the weight vector in Eq. (4) to calculate the composite index. Fig. 5 reports the annual global production proportions of five crops from 2016, showing very stable trends, and thus, we can utilize the averages of all years as final weights. The composite vulnerability indicator of all agricultural commodities is displayed in Fig. 6. The Markov Switching model is applied to model the non-linear dynamics of regime changes, and the estimated smooth probabilities of high-vulnerability states are also shown. The results suggest that the global agricultural commodity market entered a low-vulnerability era broadly from 2009. However, during 2020–2022, CPR severely impacted the prices of global crops. From then until now, the resistance to CPR has gradually become stronger.

4. Conclusions

This paper introduces a novel framework assessing the price vulnerability of global agricultural commodities to climate physical risks, highlighting critical food security implications. Our key findings show pronounced crop-specific vulnerability: among the main concerned crops, maize and soybean are highly susceptible to physical risk shocks, while rice exhibits the lowest vulnerability. The overall vulnerability of global agricultural commodities acutely burgeoned during 2005–2008 and 2020–2022, driven by extreme climate events interacting with tight market conditions.

These distinct vulnerabilities necessitate targeted policies:

For maize & soybean (high vulnerability): Prioritize investments in drought/heat-resilient varieties, enhance region-specific crop insurance/risk management tools, and promote sustainable water management and geographic diversification in key production zones.

For rice (lower vulnerability): Maintain and modernize critical irrigation infrastructure supporting its resilience and monitor emerging climate threats like heat stress and salinity.

For vulnerability peaks: Strengthen integrated early warning systems combining climate and market data. Judiciously manage strategic reserves of vulnerable staples (like maize/wheat) for timely intervention during crises. Promote transparent international trade rules to avoid export restrictions that exacerbate price spikes.

This framework provides actionable insights for government departments and stakeholders to develop evidence-based policies enhancing resilience to climate risks across specific crops and during critical periods. Of course, more diverse types of risk shocks, such as geopolitical risks and export policy changes of major producing countries, could be incorporated into future research to more comprehensively scrutinize the challenges posed by multi-dimensional heterogeneous risks to agricultural sectors.

Acknowledgments

This work was supported by the National Social Science Fund of China (Grant No 23ZDA039, No 22AZD133) and the National Natural Science Foundation of China (Grant No 72273090).

Appendix 1. Robustness check

Fig. A1.

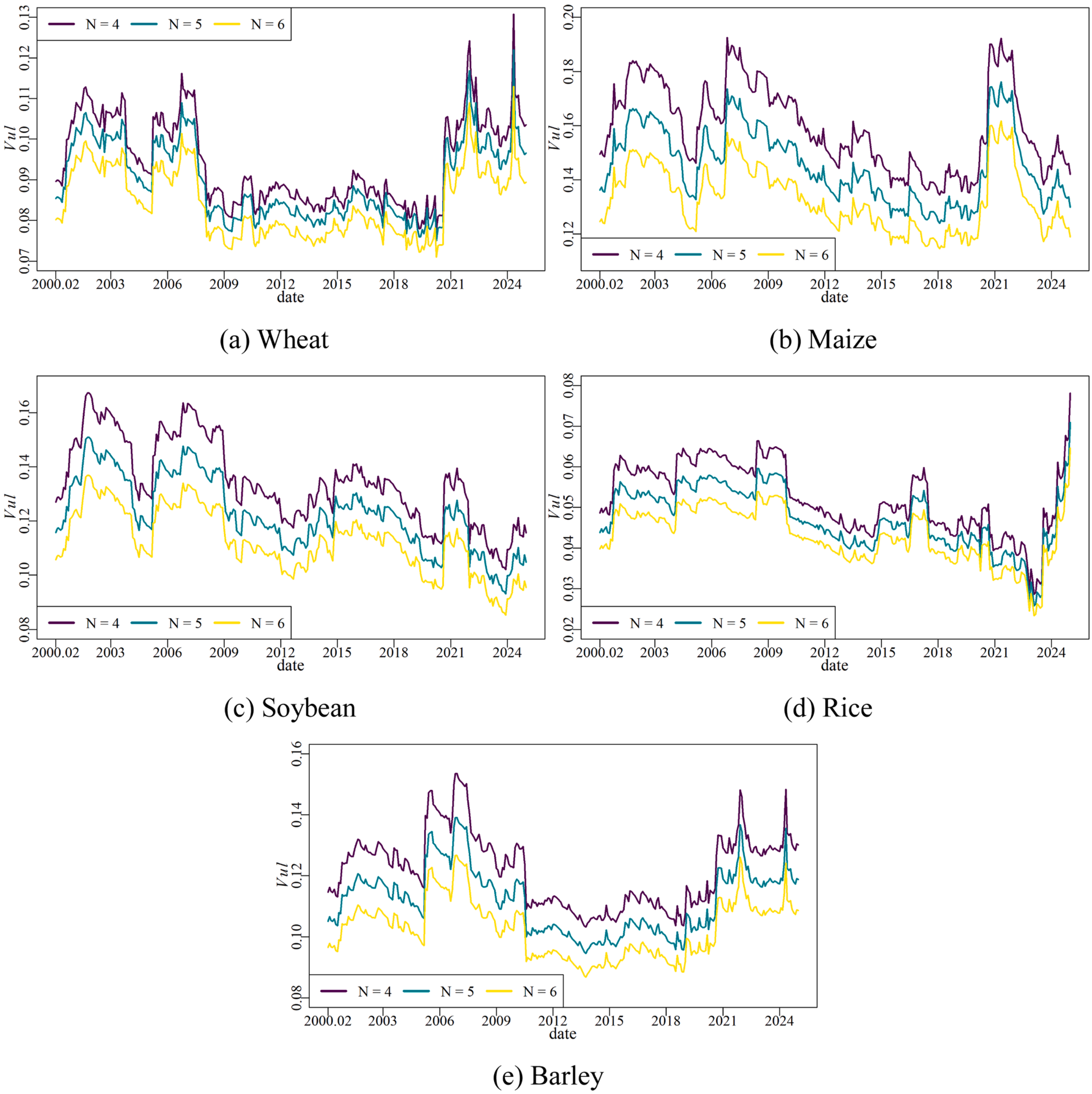


Fig. A1. Robustness tests for impulse response horizons $N \in \{4, 5, 6\}$.

Appendix 2. The time-varying cross-correlation matrix

Besides using the EWMA method to estimate Ω_t in Eq. (4), the dynamic conditional correlation (DCC) coefficient derived from the multivariable GARCH model (like DCC-GARCH) is another possible alternative. To meet the prerequisite of stationarity when estimating DCC-GARCH, the vulnerability measures of all agricultural commodities undergo log differencing. Fig. A2 shows these two composite indicators of vulnerability, suggesting their relatively similar trends with a Pearson correlation coefficient of up to 0.75.

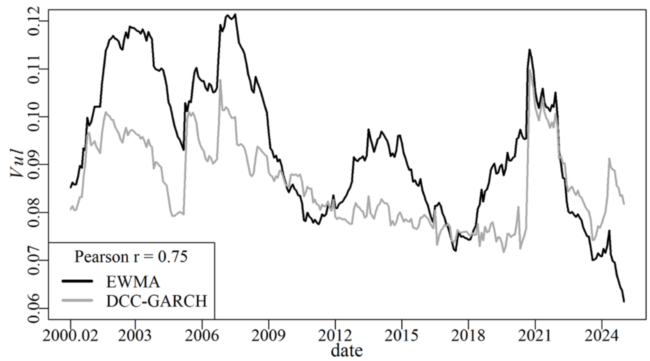


Fig. A2. Composite indicator of vulnerability with the time-varying cross-correlation matrix estimated by the EWMA or DCC-GARCH model. The Pearson correlation coefficient (0.75) between these two metrics is also reported in the figure legend.

Data availability

Data will be made available on request.

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